

Multiple Choice (10 marks)

1. Which image data augmentation is most common for natural images?
 - (a) random crop and horizontal flip
 - (b) random crop and vertical flip
 - (c) posterization
 - (d) dithering
2. As of 2020, which architecture is best for classifying high-resolution images?
 - (a) convolutional networks
 - (b) graph networks
 - (c) fully connected networks
 - (d) RBF networks
3. Averaging the output of multiple decision trees helps __.
 - (a) Increase bias
 - (b) Decrease bias
 - (c) Increase variance
 - (d) Decrease variance
4. In Yann LeCun's cake, the cherry on top is
 - (a) reinforcement learning
 - (b) self-supervised learning
 - (c) unsupervised learning
 - (d) supervised learning
5. Statement 1| A neural network's convergence depends on the learning rate. Statement 2| Dropout multiplies randomly chosen activation values by zero.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The joint probability $P(H, U, P, W)$ can be expressed as $P(H) * P(W) * P(P | W) * P(U | H, P)$.
7. True or False: The method used to learn weights, such as matrix inversion or gradient descent, greatly affects the trade-off between underfitting and overfitting.
8. True or False: As of 2020, some models achieve accuracy greater than 98% on the CIFAR-10 dataset.
9. True or False: The variance of the Maximum A Posteriori (MAP) estimate is the same as that of the Maximum Likelihood Estimate (MLE).
10. True or False: An excessively large step size during training can cause the training loss to increase over time.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to shift last element to first position in the given list.
12. Write a function to find the maximum sum of bi-tonic sub-sequence for the given array.

End of Paper